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A reactive wetting strategy improves lithium metal reversibility

Peichao Zou¹, Chunyang Wang¹, Jiayi Qin¹, Rui Zhang, Huolin L. Xin  [Show more](#)  Outline |  Share  Cite<https://doi.org/10.1016/j.ensm.2023.02.017> [Get rights and content](#) [Full text access](#)

Abstract

The cycling of lithium metal (Li^0) anode is often accompanied by irregular lithium plating/stripping, leading to poor Li^0 reversibility, and posing a grand challenge to constructing reliable Li^0 batteries. Hereby, we report a reactive wetting strategy to enable conformal and highly reversible Li^0 plating/stripping through a lithium-reactive substrate. The reactive wetting, smooth nucleation, and conformal growth behaviors of Li^0 are visualized through in-situ transmission electron microscopy, ex-situ scanning electron microscope, and in-situ optical microscope characterizations. Based on the reactive wetting substrate, high average Coulombic efficiencies (99.0% at 0.5 mA cm^{-2} for 400 cycles) are realized for Li^0 anodes. By pre-plating Li^0 into the reactive wetting substrate as anodes and leveraging LiFePO_4 as cathodes, high capacity retention (76.0% after 200 cycles in low N/P ratio conditions) is also achieved in the as-constructed Li^0 full cells. This work establishes a fundamental understanding of reactive wetting dynamics of Li^0 and provides a drop-in solution to improving Li^0 reversibility.



Keywords

Reactive wetting; Lithium metal anode; In-situ TEM

1. Introduction

Lithium metal (Li^0) battery technology offers a tantalizing solution to overcoming the energy density limitation of state-of-the-art lithium-ion batteries, owing to the high theoretical capacity and energy density merits of Li^0 anode. In particular, Li^0 is an indispensable component in Li-S and Li- O_2 batteries that promise the accomplishment of the 500 Wh/kg target based on the whole battery level [1], [2], [3]. The past decade has witnessed the reviving of this once-abandoned Li^0 anode. Yet until today, how to effectively address the safety concern and low reversibility issue of Li^0 anodes still remains a tough challenge, due to the uncontrolled Li^0 evolution, dead Li formation, and dramatic volume change during cycling [4], [5], [6].

It has been well established that the metal's growth pattern is highly related to its nuclei state, where uniform and high-density nucleation is favorable for further conformal and smooth growth [7,8]. For Li^0 deposition in liquid electrolyte-based batteries, the Li^0 nuclei state is dominated by the nucleation substrate's lithiophilicity and cell operation conditions [5]. Among those, adopting three-dimensional (3D) anode substrates with high lithium ion affinity has been proven an effective strategy to homogenize lithium ion distribution and diffusion, which then helps to smoothen Li^0 nucleation and growth. Thus far, the reported lithiophilic materials include (i) carbonaceous materials with abundant surface defects [9], [10], [11]; (ii) metals like Au, Sn, Zn, Mg, Ag, and Al that can alloy with Li metal and reduce the nucleation barrier upon the subsequent Li plating [12,13]; (iii) inorganic materials such as Si and some metal compound materials (e.g., SiO_2 , NiO, CuO, CuP, MgF_2 , and ZnO) [14], [15], [16], [17], [18] that can be electrochemically lithiated and convert into mixed ionic/electronic conductors (MIECs) before Li plating. Here, the Li plating behavior is influenced by the reaction between lithiophilic materials and Li ions/metal, similar to the liquid spreading on a solid substrate that has a reaction/absorption toward the liquid, so-called reactive wetting behavior [19,20].

In particular, leveraging reactive wetting substrates that render in-situ formation of MIECs on the substrate surface can homogenize both the Li-ion flux and surface current distribution upon Li^0 plating/stripping, which has been widely used to stabilize Li^0 metal anodes in both liquid systems and solid-state systems [21,22]. To visualize the Li plating/stripping behavior on these MIECs-forming reactive-wetting substrates, in-situ characterization tools such as in-situ optical microscopy and in-situ transmission electron microscopy (TEM) have been widely adopted. Compared with the in-situ optical observation tool, in-situ TEM technique has higher imaging resolution and can track the morphology evolution, ionic transport, electrochemical reactions, and phase transformations of various electrode materials in real time [6,17,18,23,24]. Yet, more efforts have been made to elucidate the conversion mechanism rather than the subsequent Li deposition behavior. In addition, from the technical perspective, the sample architecture used in many studies makes it hard to identify where the deposited Li is in TEM projection imaging, preventing us from directly visualizing or decoupling the reaction-mediated lithiation from the subsequent Li metal deposition process.

Here, we reported a surface sulfurized copper nanowire (S-CuNW) film as a new reactive wetting substrate for Li metal anode. Owing to the large aspect ratio of the nanowire architecture, a two-step lithiation pathway was directly observed upon Li metal deposition on single S-CuNW through the in-situ transmission electron microscopy (TEM) experiment, which has not been reported before. Coupling with in-situ optical observation, ex-situ X-ray photoelectron spectroscopy (XPS), and ex-situ scanning electron microscope (SEM), it was demonstrated that the surface CuS thin layer was firstly transformed into a mixed ionic/electronic conductor layer, and then uniform Li^0 nucleation and conformal Li^0 growth occurred upon Li plating. Electrochemical evaluation manifested that leveraging the reactive wetting substrates can improve both the Li^0 cycling reversibility and lifetime. Our work offers new insights into the Li plating behavior on reactive wetting substrates and may inspire new Li anode designs.

2. Experimental

2.1. Fabrication of CuNW

To synthesize the Cu nanowire (CuNW), a precursor solution was first prepared by dissolving 700 mg CuCl_2 (Aladdin) and 2.65 g tetradecylamine (Aladdin) into 250 mL deionized water and stirred for 24 h. Then, 1 g glucose was added into the solution and stirred for another 10 min. After that, the solution was kept at 100 °C for 8 h to grow CuNWs. All the above steps were realized

under an Ar-protected atmosphere. Finally, the as-prepared Cu NWs were washed with isopropyl alcohol by centrifugation and dried by lyophilization.

2.2. Sulfurization of CuNW and Cu foil

The sulfurization of CuNW was realized via a hydrothermal method. Briefly, 14.4 mg $\text{Na}_2\text{S}\cdot 9\text{H}_2\text{O}$ was dissolved in a mixture solution of 30 mL anhydrous ethanol and 30 mL deionized water by constant stirring. After that, the dried Cu NWs powders were added to the solution. The reaction mixture was transferred into a 100 mL autoclave and heated at 100°C for 5 h. The sulfurized Cu NW (S-CuNW) was washed with deionized water by centrifugation and dried by lyophilization.

To sulfurize Cu foil and obtain a thin CuS layer on the Cu surface, we leveraged a chemical vapor deposition (CVD) method that has been previously reported for sulfurizing Zn metal foil [25]. In this protocol, Cu foil was transferred into the heating area of a tube furnace. Pure CS_2 was stored in a single-neck flask, and then vaporized and pumped into the heating area from the opposite direction. The pure N_2 as carrier gas was filled in and the tube was pumped to less than 20 Pa for 10 min. Then open the valves for N_2 and CS_2 to maintain $\text{N}_2=8$ sccm, and $\text{CS}_2=2$ sccm. The heating temperature was kept at 250°C for 1h to obtain the sulfurized Cu foil.

2.3. Material characterizations

The SEM experiment was conducted through a scanning electron microscope (LEXI-FEI Magellan 400). X-ray photoelectron spectroscopy profiles of S-CuNW were recorded on a Kratos AXIS-Supra, where the data were calibrated using the C-C bond signal at 284.8 eV and fitted using the XPS Peak software. The transmission electron microscopy experiments were performed with an FEI Talos F200X electron microscope operated under 200 KeV. The *in-situ* TEM experiments were performed in bright-field (BF)-TEM mode with a scanning tunneling microscopy (STM)-TEM set-up. The energy-dispersive spectroscopy experiments were performed in high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) mode with a Super-X EDS detector. In situ optical observations were carried out to visualize the lithium deposition behavior on bare Cu foil and S-CuNW substrate, using a polytetrafluoroethylene cell with transparent quartz window (Kejing, STC-Q). During Li deposition process, a digital camera (Sony NEX6) was used to collect photographs with a capture interval time of 30 s. The applied current density was 2 mA cm^{-2} by using an electrochemical working station (NEWARE).

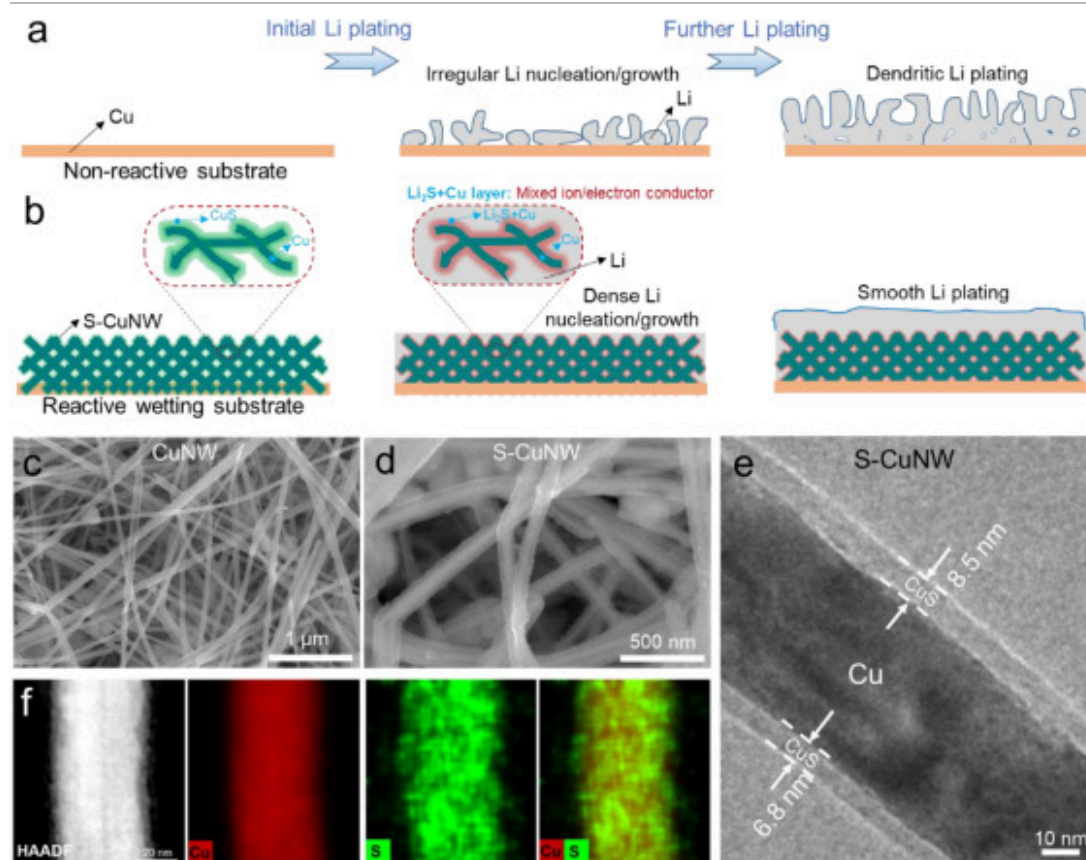
2.4. Electrochemical tests

To evaluate the S-CuNW and CuNW in stabilizing Li^0 evolution, standard CR2032 coin-type cells were used and assembled. Prior to the cell assembly, S-CuNW and CuNW film electrodes were prepared by a traditional slurry-casting method, in which S-CuNW (or CuNW), carbon black (Super-P), and polyvinylidene difluoride (PVDF) with a weight ratio of 8:1:1 was mixed together in 1-methyl-2-pyrrolidinone (NMP), followed by casting on Cu foil. The thickness of S-CuNW and CuNW film electrode was around 10 μm . The symmetric Li//Li cells were assembled using 450 μm -thick Li^0 foils as both the working electrode and the counter/reference electrode. The asymmetric Li//Cu cells were constructed using 450 μm -thick Li^0 foils as the counter/reference electrode and Cu^0 foil (12 μm in thickness) as the working electrode. All these cells were prepared in an Ar-filled glove box using Celgard-2500 PP separators and a traditional ether-based electrolyte (1 M LiTFSI in DOL/DME, 1:1 in vol., with 2wt.% LiNO_3). Galvanostatic discharge/charge (GCD) profiles were obtained from a NEWARE electrochemical testing system. For the low N/P ratio LFP//S-CuNW@Li and LFP//Cu foil@Li full cells test, the LFP (LiFePO_4) cathode was prepared using a similar slurry-casting method, where the mass ratio between LFP, Super-P, and PVDF was controlled at 8:1:1. The slurry was thereafter casted on Al foil followed by vacuum drying overnight at 80 $^{\circ}\text{C}$. To realize the low N/P ratio, the Li anode was fabricated by electro-plated a designated amount of Li^0 on Cu foil and S-CuNW film electrode using the asymmetric cell configuration, using the conventional ether-based electrolyte. The Li//LFP-type full cells were cycled at the voltage window of 2.7–4.2V (vs. Li^+/Li) to investigate the cycling performance.

3. Results and discussions

To obtain the sulfurized Cu nanowire substrate, bare Cu nanowires were firstly fabricated through a simple wet [chemical method](#) [14]. Thereafter, the sulfurization of Cu nanowire was realized by a modified [hydrothermal method](#), in which Na_2S was used as the sulfurization source. Note that the hydrothermal reaction time should be controlled to obtain the surface CuS layer with a moderate thickness, since too thick CuS layer will build up the [electron transfer](#) resistance while too thin CuS layer will lead to insufficient smoothening effect of Li [metal deposition](#). In this work, the optimized S-CuNW was selected with the hydrothermal reaction of 5 h, which was used as the model system to demonstrate the reactive wetting concept. Scanning electron microscopy images show that the nanowire morphology was well maintained after sulfurization (Fig. 1c, d). TEM image shows that the surface CuS sheath on the Cu core has a thickness varying from 6.8 nm to 8.5 nm (Fig. 1e). The energy-dispersive X-ray spectrometer (EDS) mappings in TEM suggests that the S element is dominantly distributed on the nanowire surface (Fig. 1f), further proving the core-shell structure. X-ray photoelectron spectroscopy (XPS) spectra of Cu 2p and S 2p also validate the successful surface sulfurization of CuNW (Fig. 2a, b). The Cu 2p $1/2$ peak at 952.8 eV and 2p $3/2$ peak at 932.8 eV can be indexed

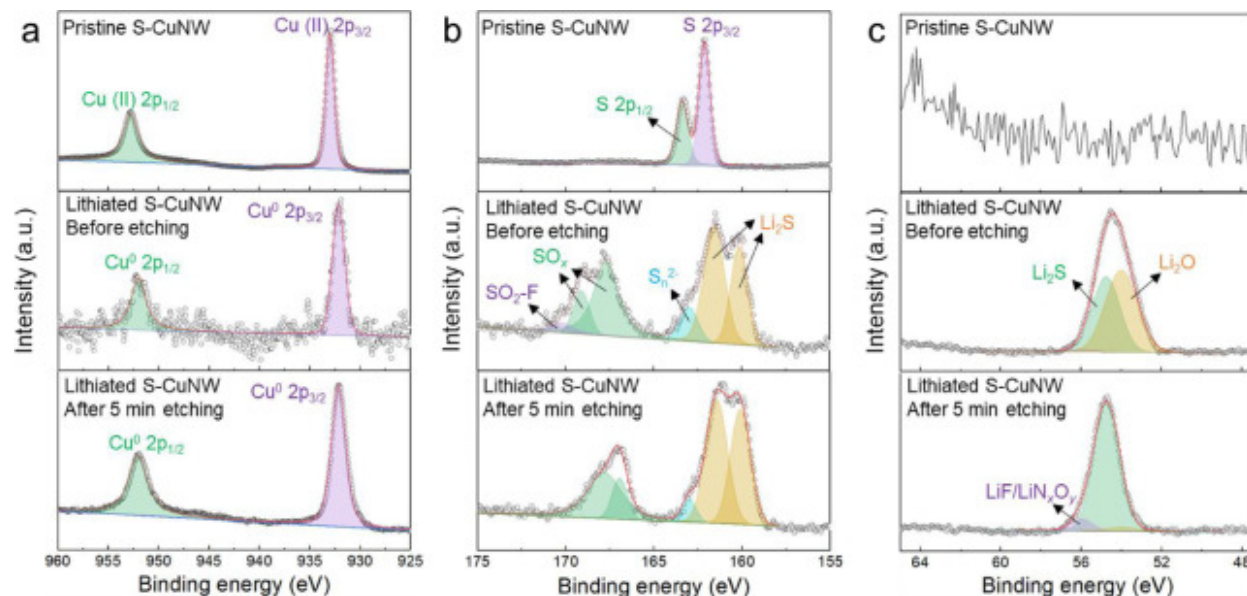
to the Cu-S bonding. Correspondingly, the characteristic S 2p 1/2 peak at 163.3 eV and 2p 3/2 peak at 162.2 eV is attributed to the existence of surface CuS [26].



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Fig. 1. Schematic comparison of the different Li plating behaviors on (a) non-reactive substrate (Cu foil) and (b) reactive-wetting substrate (S-CuNW). SEM images of (c) Cu nanowire (CuNW) and (d) sulfurized Cu nanowire (S-CuNW). (e) TEM image of single S-CuNW demonstrating the CuS sheath on Cu core. (f) EDS mappings showing the Cu and S elements distribution within S-CuNW.



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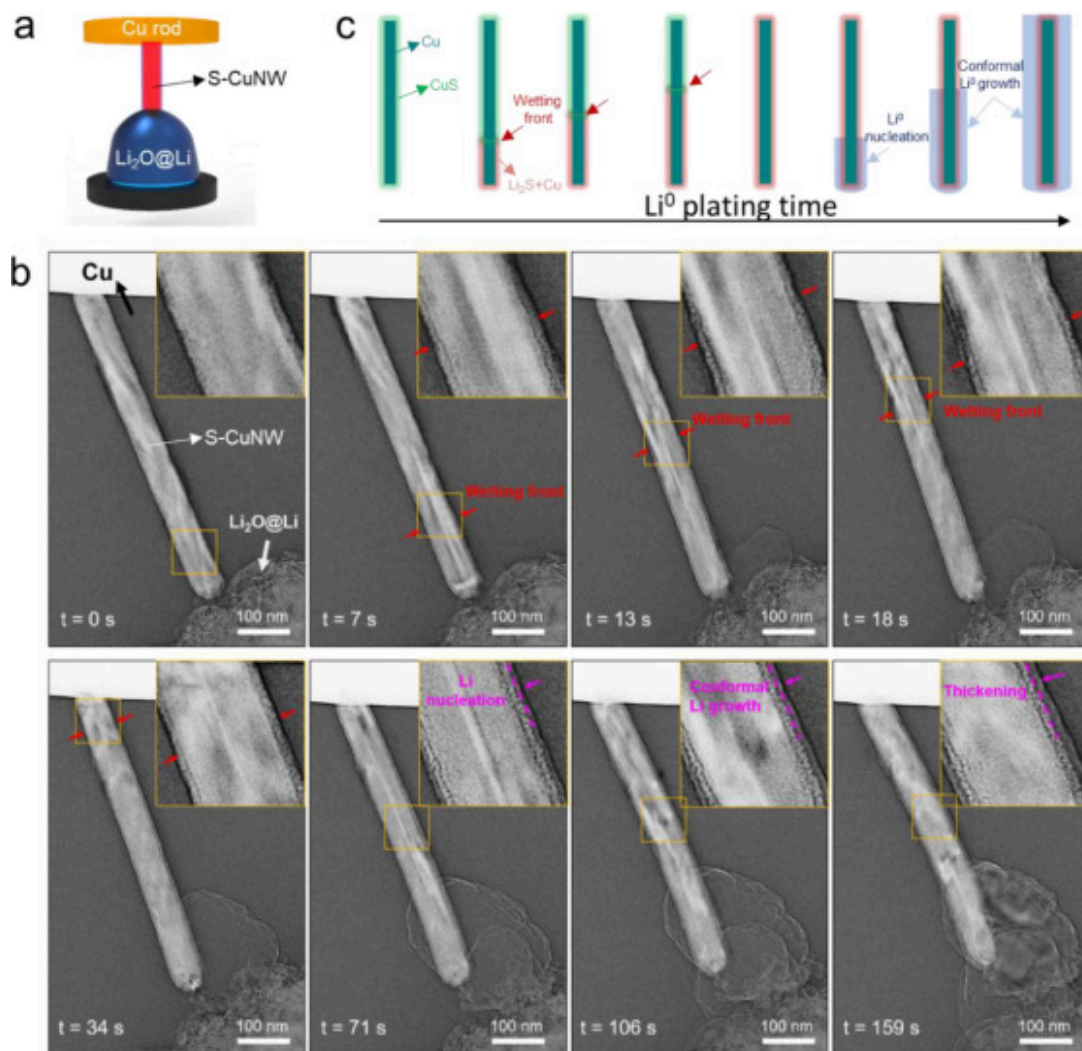
Fig. 2. (a) Cu 2p, (b) S 2p, and (c) Li 1s XPS spectra of S-CuNW before and after lithiation. The lithiation process was realized by discharging the Li//S-CuNW half-cell from open circuit voltage to 0.01V (vs. Li⁺/Li).

Since S-CuNW was used as the substrate of Li metal anode, the final surface chemistry of S-CuNW before Li deposition is critical to the Li metal nucleation and grow behaviors. Hereby, we further leveraged XPS to analyze the chemistry information of lithiated S-CuNW, which was fabricated by discharging the material from open circuit potential to 0.01 V (vs. Li⁺/Li) in an asymmetric Li//S-CuNW cell configuration. The XPS measurement shows that the Cu⁰ signal emerged along with the disappear of Cu (II) signal in the lithiated S-CuNW, particularly after etching (Fig. 2a). Simultaneously, Li₂S was observed on the surface of lithiated S-CuNW (Fig. 2b, c). This is because of the transformation of CuS layer into Li₂S and Cu⁰ upon lithiation, following the two-step reaction as shown below [27]: $\text{CuS} + \text{Li}^+ + \text{e}^- \rightarrow 1/2 \text{Cu}_2\text{S} + 1/2 \text{Li}_2\text{S}$ and $1/2 \text{Cu}_2\text{S} + \text{Li}^+ + \text{e}^- \leftrightarrow \text{Cu} + 1/2 \text{Li}_2\text{S}$. The SO_x, S_n²⁻, SO₂-F species in S 2p spectra and Li₂O, LiN_xO_y species in Li 1s spectra were derived from the solid-electrolyte interphase (SEI) layer on the surface of lithiated S-CuNW, whose signal significantly diminished after etching. The speculated lithiation process was also in accordance with the cyclic voltammetry result (Fig. S1), in which a sharp cathodic peak centered at around 1.5 V (vs Li⁺/Li) is related to conversion reaction of Li storage in CuS [28]. The broad weak peak at 0.5-0.8 V corresponds to the formation of SEI

layer during the initial discharge process. Upon the subsequent charge process, two peaks at around 1.9 and 2.3 V are due to the Li-ion extraction from the lithiated S-CuNW, which agrees well with previous reports [27,28].

As discussed above, prior to Li metal deposition, the CuS sheath will first convert to a mixed ion/electron conductive (MEIC) layer containing Li_2S and Cu due to the electrochemical lithiation of CuS. This process ensures the preferential absorption of lithium ions around CuS layer, since the CuS layer functions as the electrochemical reactive material here and it has better lithiophilicity than metallic Cu (inactive material). In the as-formed MEIC layer, the high Li^+ conductivity of Li_2S and electronic conductivity of Cu phase can further homogenize the Li^+ flux and surface current distribution, thus rendering smooth Li nucleation/growth. The formation of Li_2S and Cu renders a MIEC layer on the surface of lithiated S-CuNW, which is expected to homogenize the Li^+ flux and surface current distribution, thus smoothening Li^0 plating/stripping [22].

To further explore the Li plating behavior on S-CuNW, an in-situ Li plating experiment in a TEM was conducted (Fig. 3 and Movie S1). Fig. 3a shows the schematic illustration of the TEM setup for the in-situ Li deposition. To perform the experiment, an S-CuNW was inserted between a Cu electrode and Li metal (a layer of lithium oxide was formed on the surface of the Li metal and it acts as a Li^+ conductor for the Li deposition). Fig. 3b presents time-resolved bright-field (BF)-TEM images showing the in-situ Li deposition dynamics under an applied bias (-5V). It is seen that upon lithiation ($t=0$ s), reactive wetting (the wetting fronts are indicated by red arrows) soon nucleated from the bottom of the nanowire through conversion reaction which results in the decomposition of the CuS sheath. With prolonged lithiation (from $t=7$ s to 34 s), the reactive wetting front continued to propagate up along the surface of the nanowire. Later on, aided by the wetting layer, a conformal thin layer of Li metal (the interface between the reactive wetting layer and the deposited Li metal is denoted by the magenta dash line) started to grow out of the conversion-reacted layer ($t=71$ s) and the thickness of the Li metal continued to be increased uniformly ($t=159$ s), without formation of dendrites. These results indicate that the reactive-wetting substrates are favorable for smooth Li^0 nucleation and further conformal Li^0 growth. The wetting, nucleation, and growth pathway, which is directly observed by TEM for the first time, is schematically illustrated in Fig. 3c.



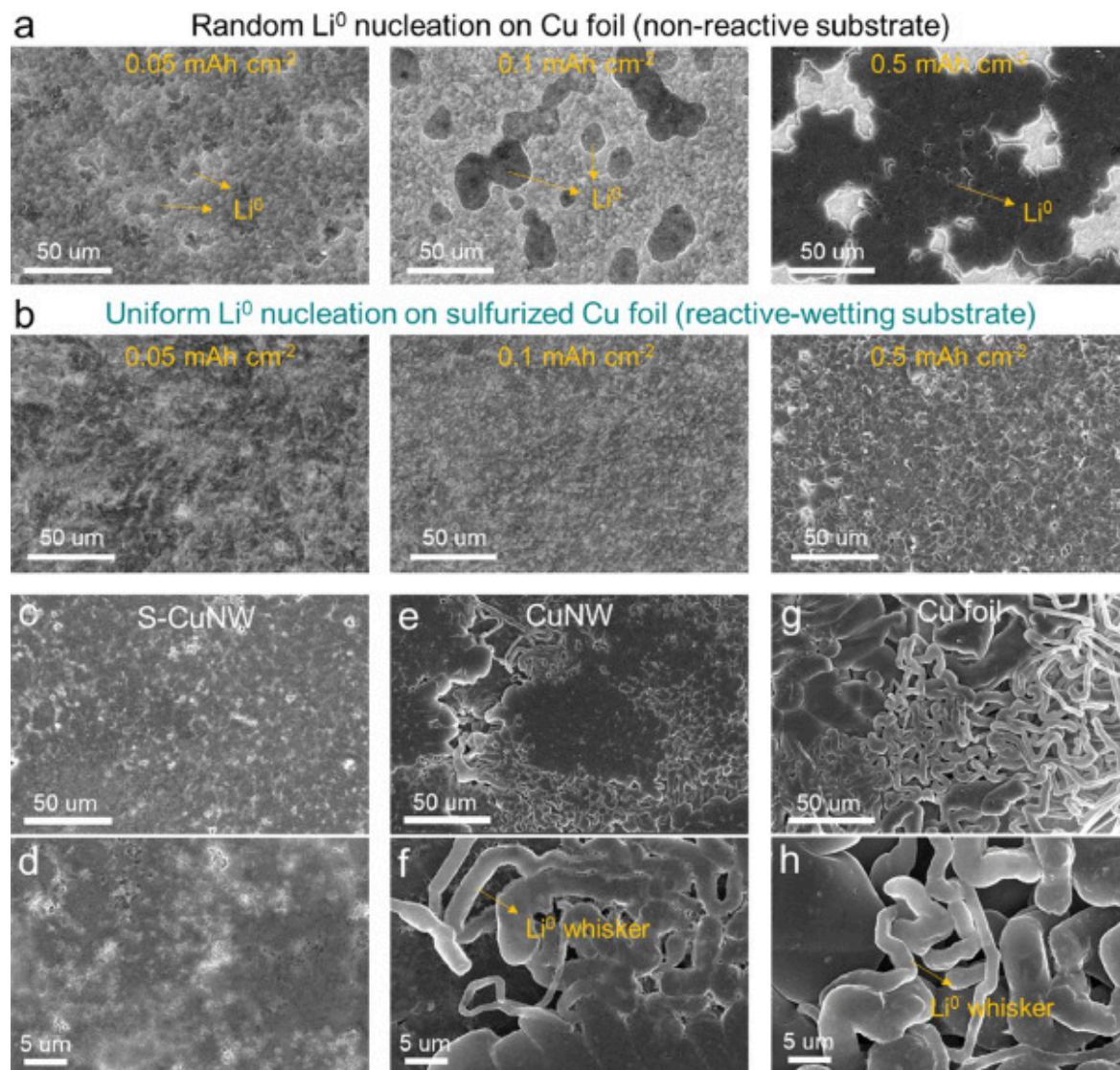
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Fig. 3. (a) Schematic illustration of the experimental setup for in-situ electrochemical Li plating in TEM. (b) Time-resolved bright-field (BF) TEM images (contrast inverted) of an S-CuNW showing the reactive wetting, Li⁰ nucleation, and conformal Li⁰ growth during the initial Li plating from t = 0 s to t = 159 s. Insets show the zoom-in images of the reaction or Li growth fronts. The reactive wetting fronts are indicated by red arrows. The interfaces between the reactive wetting layer and the deposited Li metal

are denoted by the magenta dash lines. (c) Schematic illustration of reactive wetting, Li^0 nucleation, and conformal Li^0 growth on the S-CuNW.

Aside from the in-situ TEM observation, the Li^0 nucleation and deposition behavior on reactive substrates (sulfurized Cu foil and Cu nanowires) was further investigated in 2032-type coin cells by using the conventional ether-based electrolyte (1 M LiTFSI in DOL/DME, 1:1 in vol., with 2wt.% LiNO_3). Sulfurized Cu foil, compared with bare Cu foil, was first used to demonstrate the Li nucleation process. It was found that irregular and randomly distributed Li^0 nuclei were formed on bare Cu (Fig. 4a); comparatively, uniform Li metal was nucleated on sulfurized Cu foil (Fig. 4b). This is consistent with the in-situ TEM analysis (Fig. 3) that Li-reactive layer on Cu surface is beneficial for uniform Li metal nucleation. Thereafter, S-CuNW thin film was fabricated (see experimental for details) to explore the Li deposition behavior under larger capacities. As shown in Fig. S2, the as-fabricated S-CuNW and CuNW film electrodes deliver a smooth surface morphology with S-CuNW (or CuNW) uniformly distributed with carbon black. The thickness of S-CuNW film was around 8 μm (Fig. S4c). After plating 0.5 mAh cm^{-2} (Fig. S3a, b) and 3 mAh cm^{-2} (Fig. 4c, d) of Li^0 on S-CuNW, smooth surface morphology is retained without obvious protrusions, indicating the uniform Li^0 plating behavior. In comparison, less regular Li^0 domes can be found on CuNW film electrode and Cu foil after plating 0.5 mAh cm^{-2} of Li^0 (Fig. S3c-f). Worse more, Li^0 whiskers were identified on bare Cu foil surface and CuNW film electrode after plating 3 mAh cm^{-2} of Li^0 (Fig. 4e-h). Even after Li^0 plating/stripping for 20 cycles with both a low capacity of 0.5 mAh cm^{-2} and a relatively high capacity of 3 mAh cm^{-2} , S-CuNW film electrode still maintained a smooth and dendrite-free morphology, while dendritic Li evolved on CuNW film electrode and Cu foil (Fig. S4 and S5). The smooth Li deposition behavior on S-CuNW can be attributed to the reactive wetting property of S-CuNW that can improve the surface lithiophilicity and guide uniform Li^0 plating.

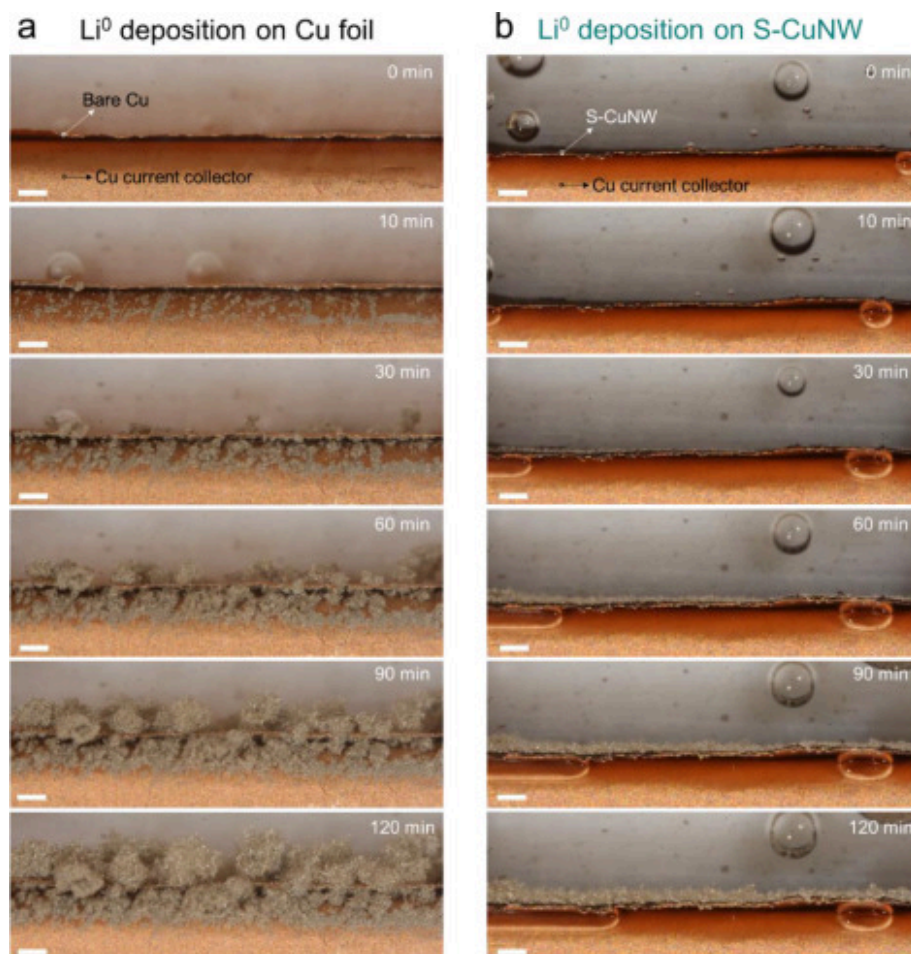


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Fig. 4. SEM images of Li^0 nucleation on (a) bare Cu foil and (b) sulfurized Cu foil at different capacities: 0.05 mAh cm^{-2} (left); 0.1 mAh cm^{-2} (middle); 0.5 mAh cm^{-2} (right). Top-view SEM images of (c, d) S-CuNW, (e, f) CuNW, and (g, h) Cu foil after plating 3 mAh cm^{-2} of Li^0 . The Li^0 plating current density in (a-h) was 0.05 mA cm^{-2} .

The selective and smooth Li metal deposition on S-CuNW was further confirmed by the in-situ optical observation experiment. Fig. 5 depicts the morphological evolution of deposited Li metal on bare Cu foil and S-CuNW substrates at 2 mA cm^{-2} . As can be seen in Fig. 5a and Movie S2, Li metal was deposited on both the working electrode (bare Cu) and current collector (Cu foil), which evolved into mossy and irregular Li after deposition for 120 mins. On the contrary, when using S-CuNW as the Li plating substrate (Fig. 5b and Movie S3), we found that Li metal only forms on the surface of working electrode (i.e., S-CuNW) while there are no Li deposits on the bottom current collector (Cu foil). This strongly evidences that Li preferentially nucleates and deposits on S-CuNW rather than Cu foil, which can be attributed to the higher lithiophilicity of S-CuNW. More importantly, the deposited Li metal was uniformly covered on the surface of S-CuNW without obvious large protrusion formation during the whole 120 min-test.



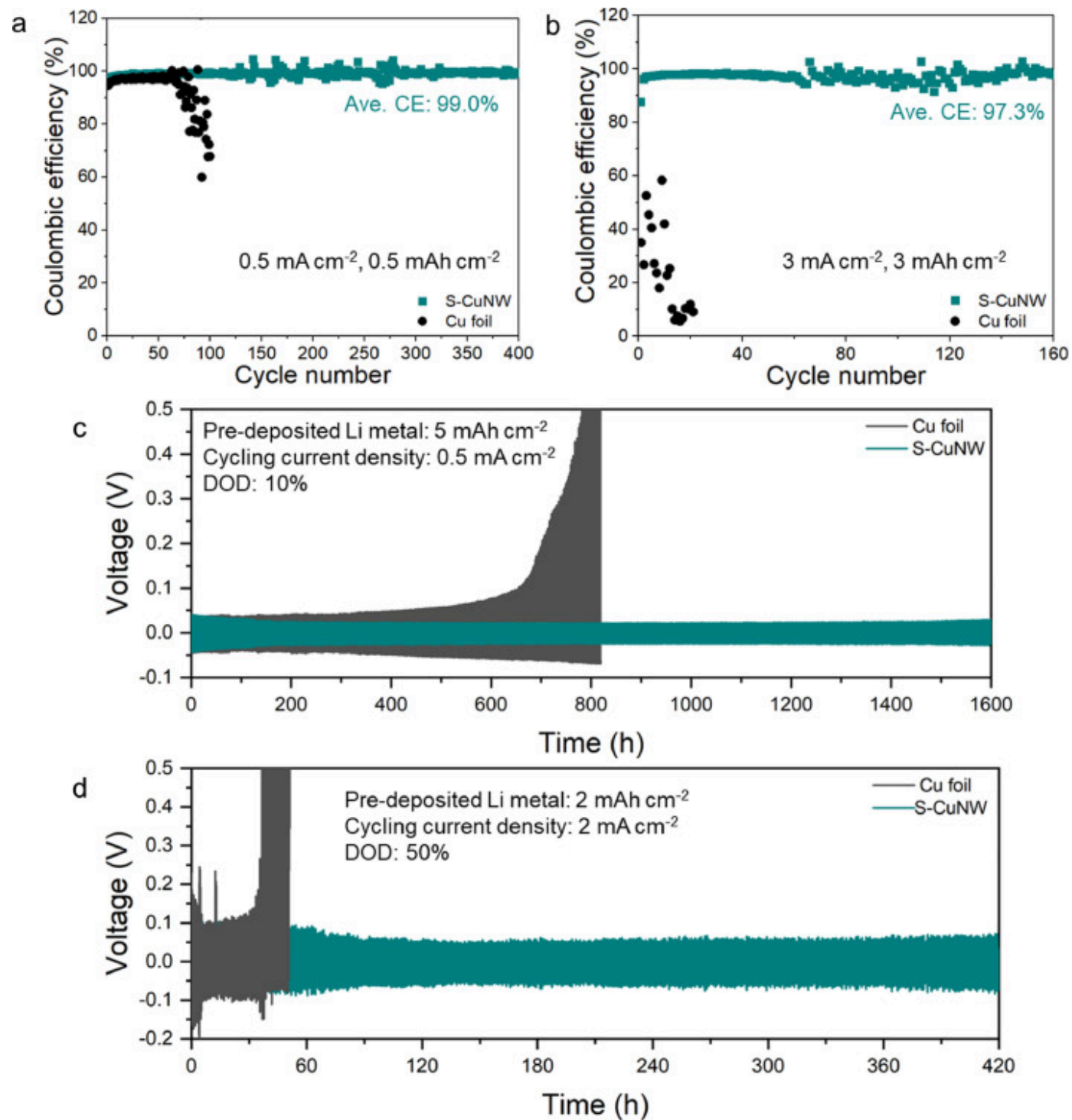
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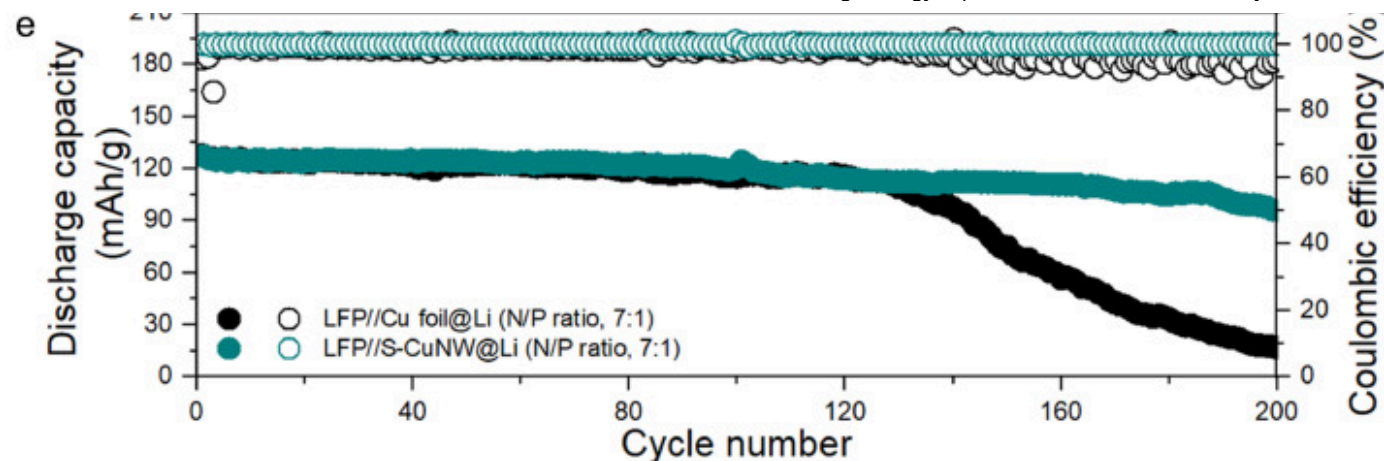
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Fig. 5. In situ optical photography observations (captured by digital camera) of the electrolyte–electrode interface during Li^0 deposition on (a) bare Cu foil and (b) S-CuNW. The deposition current density was 2 mA cm^{-2} . Scale bars in (a) and (b) are 200 μm .

To evaluate the Li^0 plating/stripping reversibility and lifetime on S-CuNW, asymmetric Li//Cu half cells were assembled and cycled under different current densities and capacities. As presented in Fig. 6a, b, S-CuNW-based cells generally show a higher average CE and longer lifetime than the control cell using Cu foil as the anode substrate. Notably, a high average CE of 99.0% and

97.3% was realized for S-CuNW-based cell when cycling at 0.5 mA cm^{-2} , 0.5 mAh cm^{-2} (for 400 cycles, Fig. 6a), and 3 mA cm^{-2} , 3 mAh cm^{-2} (for 160 cycles, Fig. 6b), respectively. In sharp contrast, Li-Cu cell based on bare Cu foil shows a lower CE and fast decay, which was shorted after only 20 cycles at 3 mA cm^{-2} , 3 mAh cm^{-2} (Fig. 6b). In addition, the reversibility of Li plating/stripping on different substrates was also evaluated in symmetric cell configurations, in which a certain amount of Li metal was deposited on the substrate followed by cycling at different current densities and depth of discharge (DOD). The rate capability test shows that the S-CuNW-based Li anode could stably operate at 8 mA cm^{-2} with a high DOD of 80% (Fig. S6); in comparison, bare Cu-based Li anode showed much dramatic polarization when the current density increasing to 2 mA cm^{-2} (DOD: 20%). Aside from the rate capability test, the long-term stability of the composite anodes was also evaluated in the symmetric cells. As shown in Fig. 6c, when pre-plating 5 mAh cm^{-2} of Li metal on Cu foil, the as-constructed Li//Cu foil@Li cell can stably operate for about 800 h with a relatively low cycling current density of 0.5 mA cm^{-2} and DOD of 10%; comparatively, the Li//S-CuNW@Li cell exhibited a much longer lifetime of 1600 h. Similarly, Li//S-CuNW@Li cells delivered much better stability than Li//Cu foil@Li cells at different DOD conditions (Fig. S7). In particular, even when cycling at a relatively higher current density of 2 mA cm^{-2} and a higher DOD of 50% (Fig. 6d), the Li//S-CuNW@Li cell can still sustain a long cycle life of 420 h without cell failure while the Li//Cu foil@Li cell dramatically decayed within 50 h. The superiority of Li//S-CuNW@Li cells was primarily related to the aforementioned smooth Li evolution on reactive-wetting substrate, and the influence of SEI layer can be excluded as the SEI composition of the deposited Li metal on both S-CuNW and bare Cu foil was very similar (Fig. S8). The SEI was consisted of mixed inorganic and organic components, such as LiF, Li_2O , Li_2S , LiN_xO_y , C-C/C-H, COR and COOR species, which are prevalent in the ether-based systems [29,30].





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Fig. 6. The CEs of Li plating/stripping on S-CuNW and Cu foil under different conditions: (a) 0.5 mA cm^{-2} , 0.5 mAh cm^{-2} ; (b) 3 mA cm^{-2} , 3 mAh cm^{-2} . (c, d) Long-term cycling stability of S-CuNW and Cu foil-based Li electrodes in symmetric cells at different current densities: (c) 0.5 mA cm^{-2} , DOD 10%; (d) 2 mA cm^{-2} , DOD 50%. (e) Cycling performance of LFP//S-CuNW@Li and LFP//Cu foil@Li full cells at 1 C and with a N/P ratio of 7:1.

To verify the feasibility of reactive-wetting substrate (S-CuNW) in stabilizing Li anode in full cells, a controlled amount of Li metal was pre-deposited on S-CuNW (denoted as S-CuNW@Li) as the composite anode, which was then paired with LiFePO_4 (LFP)

cathode to construct the LFP//S-CuNW@Li full cells. By balancing the LFP cathode loading and deposited Li amount in S-CuNW, a low N/P ratio of 7:1 was achieved for all full cells. As depicted in Fig. 6e, the LFP//S-CuNW@Li full cell showed stable cycling for 200 cycles with a capacity retention of 75.4%. Comparatively, the LFP//Cu foil@Li full cell delivered relatively stable cycling within the initial 130 cycles, which however exhibited a sudden and fast capacity drop in the following cycles. The capacity retention of LFP//Cu foil@Li full cell after 200 cycles was only 12.5%. Here, the rapid failure of the LFP//Cu foil@Li full cell was mainly due to the dendritic Li deposition on bare Cu foil substrate that leads to dead-Li growth/accumulation and electrolyte consumption. Since the Li source on the anode side is limited, the LFP//Cu foil@Li full cell showed sudden capacity decay once fresh Li metal is consumed out. When leveraging S-CuNW as the Li⁰ substrate, the Li plating/stripping reversibility on the anode side is higher, and therefore, the as-constructed LFP//S-CuNW@Li full cell presented a slower capacity decay.

4. Conclusion

In summary, by coupling in-situ TEM and optical visualizing, ex-situ SEM imaging, and electrochemical evaluation, we demonstrated that reactive-wetting substrates can smoothen Li metal deposition through promoting the lithiophilic wetting, uniform nucleation, and conformal growth behaviors. High lithium metal plating/stripping reversibility, e.g., an average Coulombic efficiency of 99.0% at 0.5 mA cm⁻², was thus realized using the reactive-wetting substrates. Our work provides the first direct TEM observation of two-step lithiation for Li metal deposition on reactive-wetting substrates, which may also usher in useful guidance for the development and optimization of highly stable and reversible lithium metal anodes for next-generation lithium-metal batteries.

CRedit authorship contribution statement

Peichao Zou: Methodology, Investigation, Data curation, Writing – original draft. **Chunyang Wang:** Investigation, Visualization, Writing – original draft. **Jiayi Qin:** Investigation, Visualization. **Rui Zhang:** Investigation. **Huolin L. Xin:** Conceptualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Appendix. Supplementary materials

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